

Previous attachment

TOSI x SUS x UAEM



Open Science Research Fair

The Bellini Atrium | Feb 20, 5-7pm

Explore the practice of open science in research, connect with PIs and graduate students, and enjoy refreshments!

Office hours

REVISED

- Fridays **12-1**pm (zoom or Stewart Biology, N5/8)

<https://mcgill.zoom.us/j/81407275167?pwd=8Cgm0RPnWQRlOdaCNFyXnS3M8amQUj.1>

- By appointment
- Before or after class, right here in Leacock 132

Large Scale Chromosomal Changes

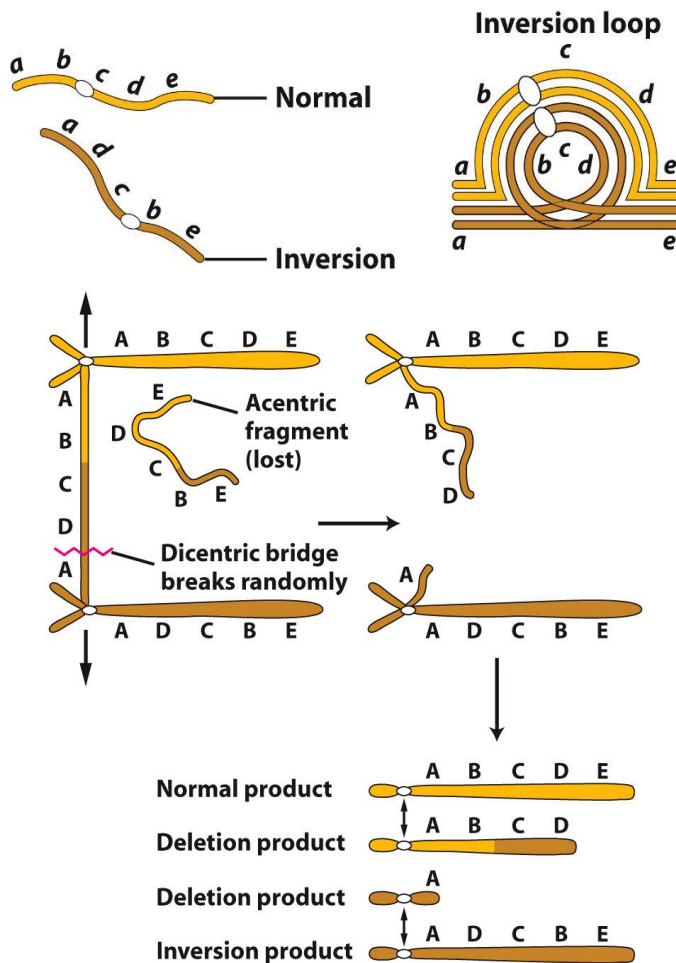
Changes in Chromosome Structure

Chromosomal inversions

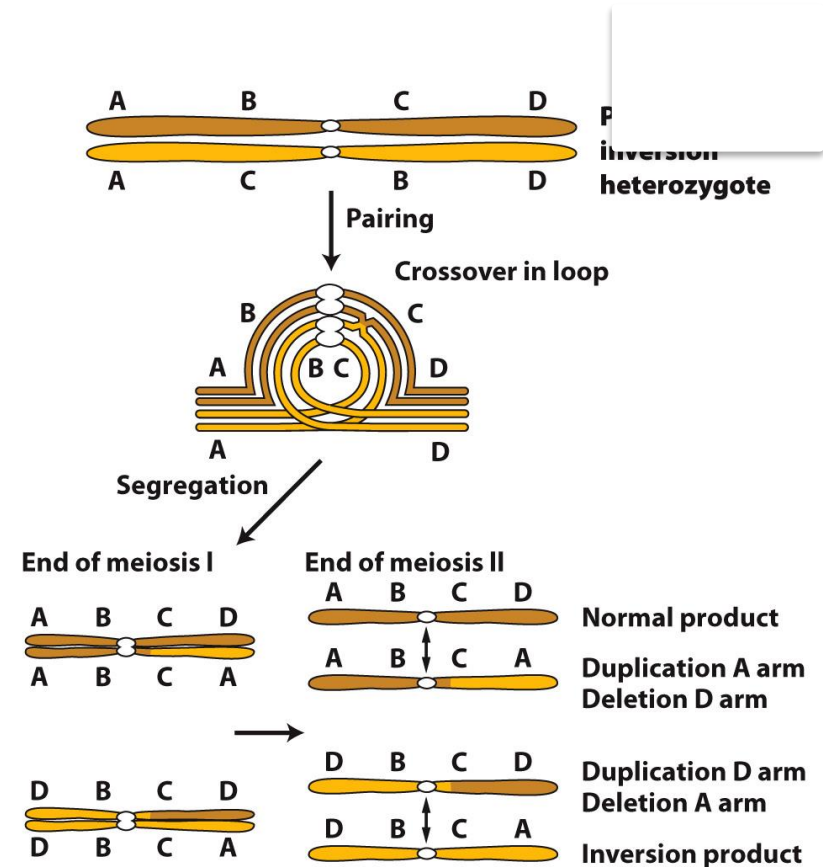
Effects on chromosome behavior during meiosis

Meiotic recombination in inversion heterozygotes leads to gametes with duplications and deletions

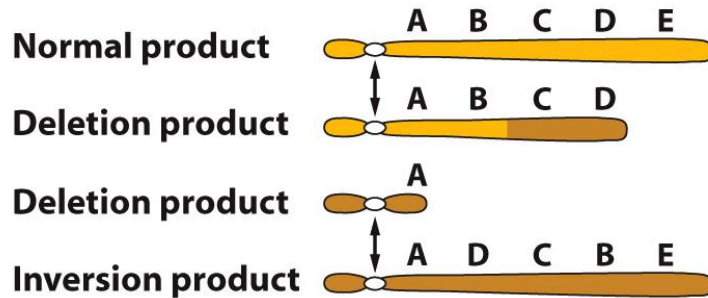
Paracentric inversion heterozygote



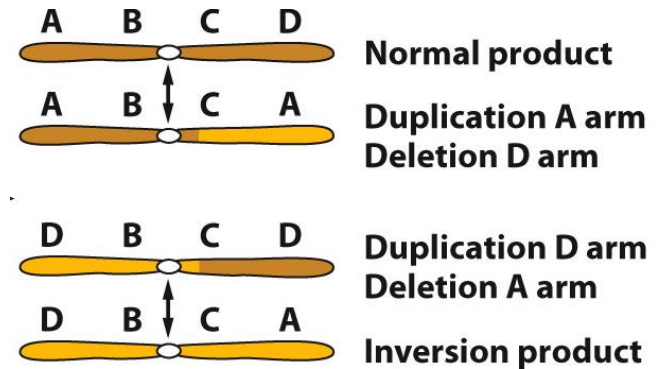
Pericentric inversion heterozygote



meiotic products: paracentric inversion



meiotic products: pericentric inversion



Think Break

What clinical consequences might you expect in an adult individual with a chromosomal inversion?

Effect on frequency of recombinant progeny

Example from *Drosophila*

dp: dumpy

cn: cinnabar

45 map units apart

$++/++ \times dp\ cn/dp\ cn$
 $+/+/dp\ cn \times dp\ cn/dp\ cn$

F1

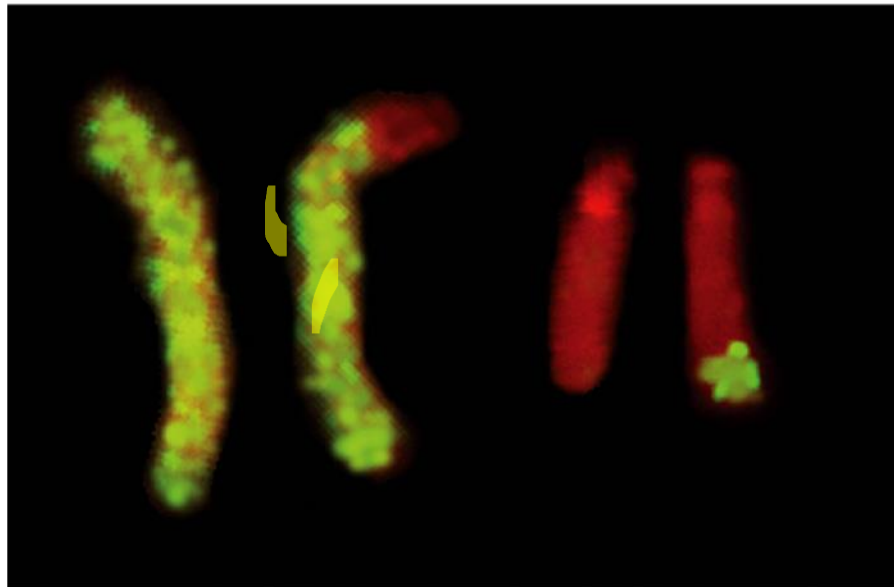
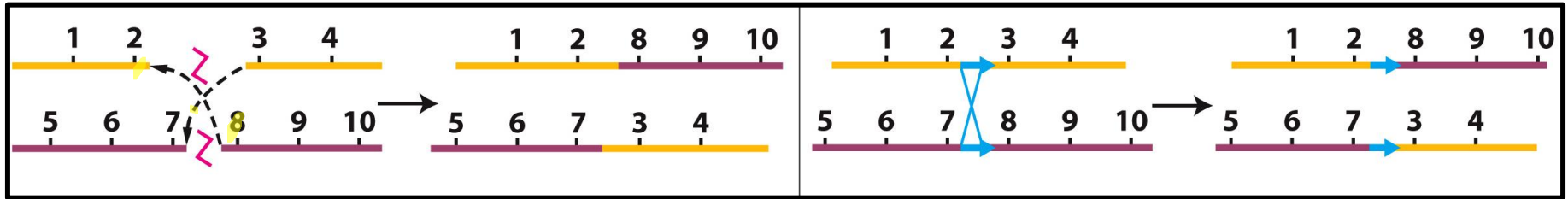
250	wild type	$+/+/dp\ cn$
246	dumpy cinnabar	$dp\ cn/dp\ cn$
5	dumpy	$dp\ +/dp\ cn$
7	cinnabar	$+ cn/dp\ cn$

F2

Expect 45%
recombinant
progeny

Germ cells with
a chromosome
that underwent
crossing over
within the
inversion did not
produce viable
progeny.

Chromosomal translocations



Chapter 17 Opener

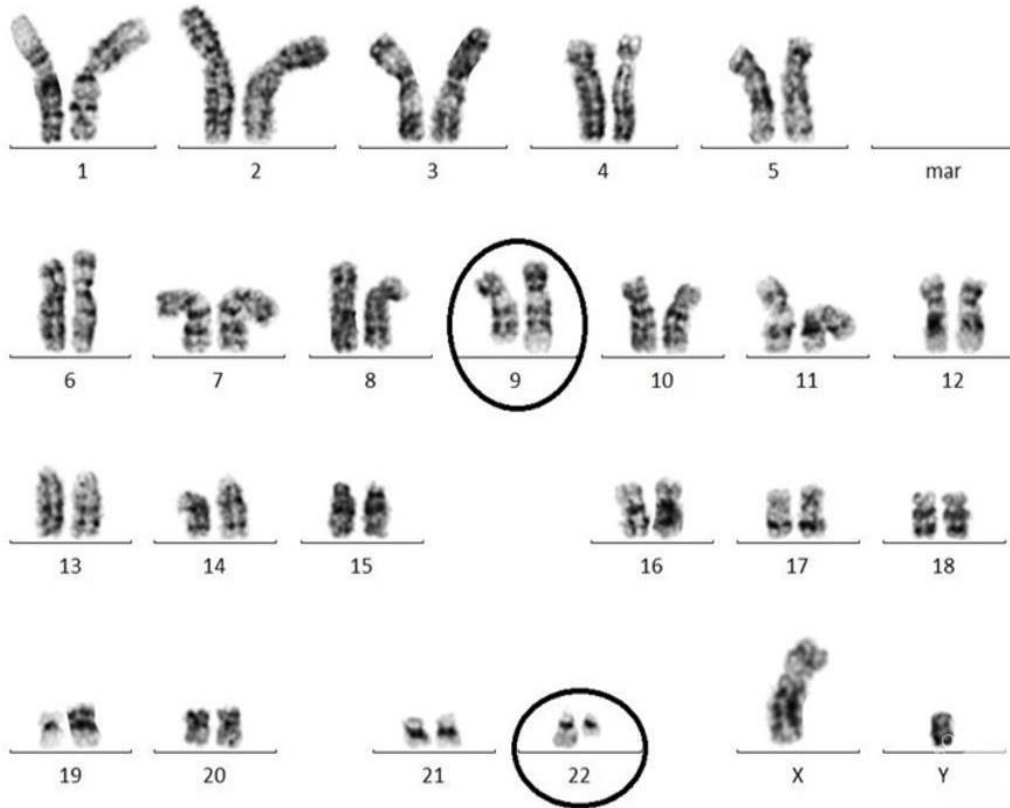
Introduction to Genetic Analysis, Eleventh Edition

Addenbrookes Hospital/Science Source

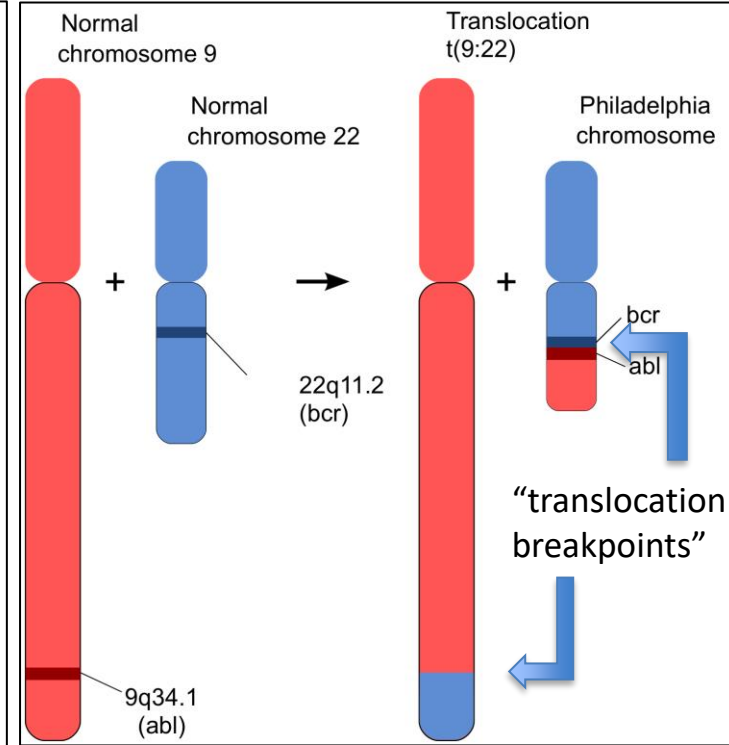
BIOL 202 W2025 | © LA Nilson

Chromosomal translocations

somatic consequences



First chromosome abnormality linked with a specific human cancer



Formation of a hybrid oncogene: chronic myelogenous leukemia

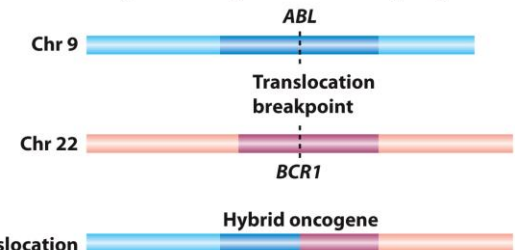
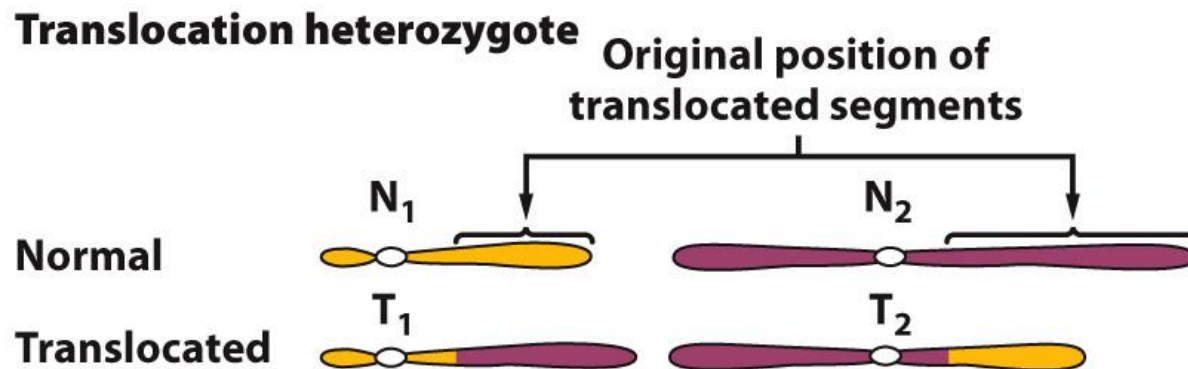


Figure 17-35b
Introduction to Genetic Analysis, Eleventh Edition
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Chromosomal translocations

somatic consequences

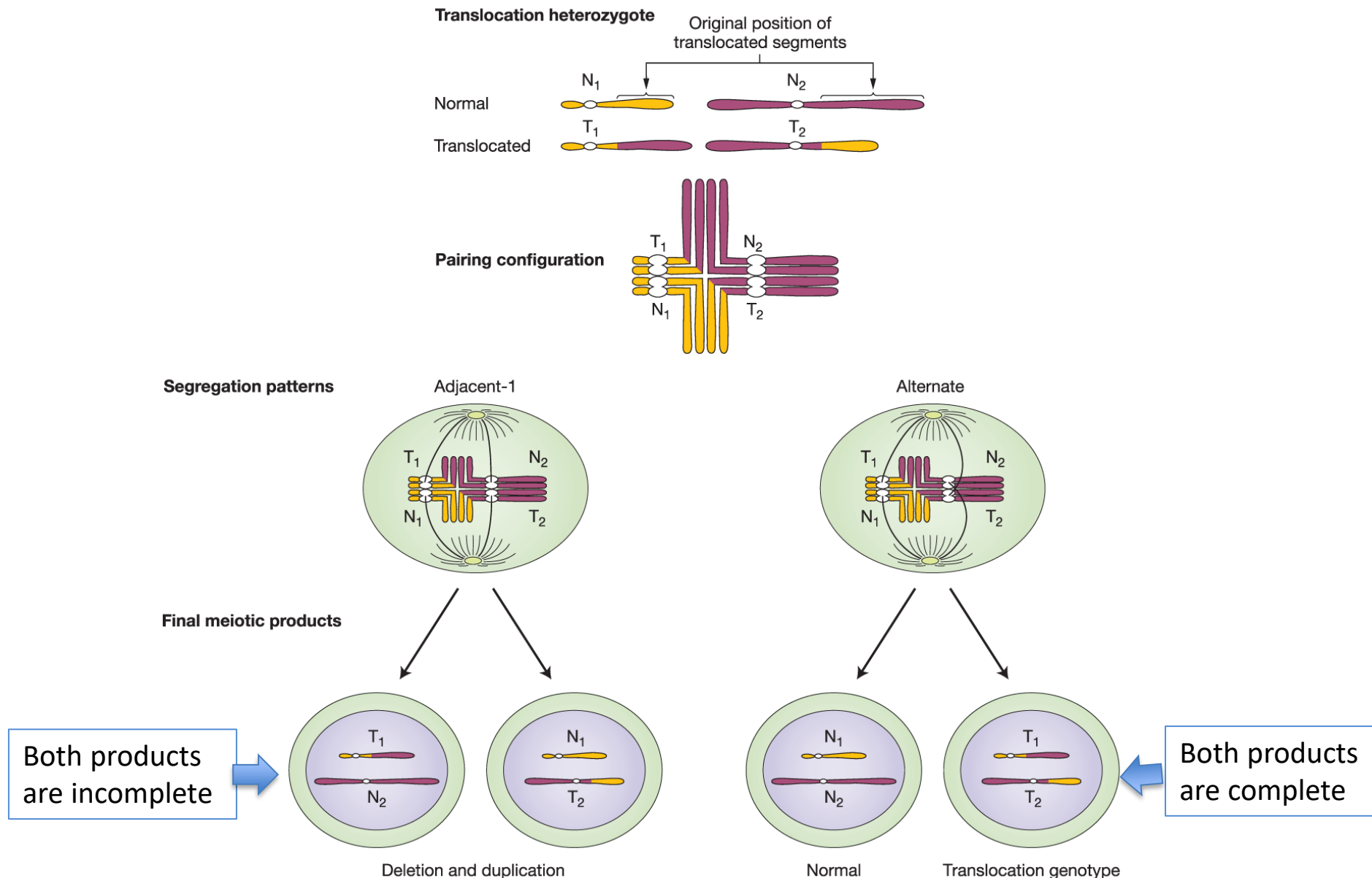
balanced translocations:



What phenotype would you predict for an individual with this translocation?

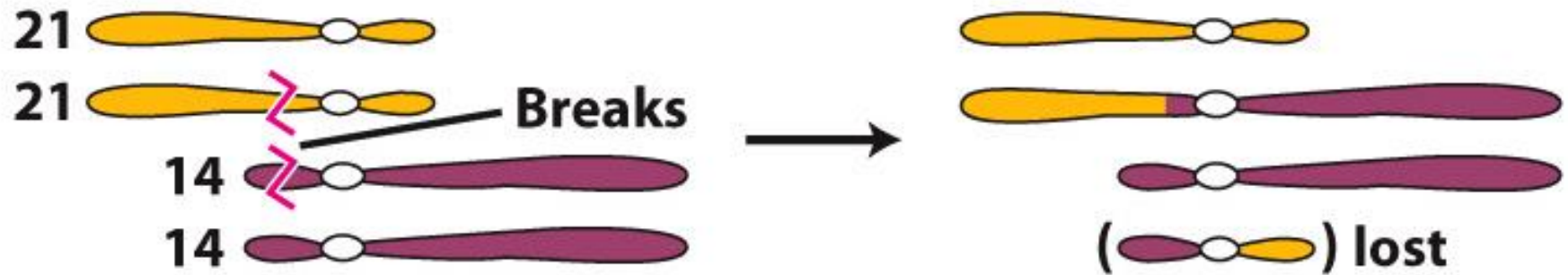
Balanced translocations and meiosis

two segregation patterns in a reciprocal-translocation heterozygote



Consequences of balanced translocations

Robertsonian translocations

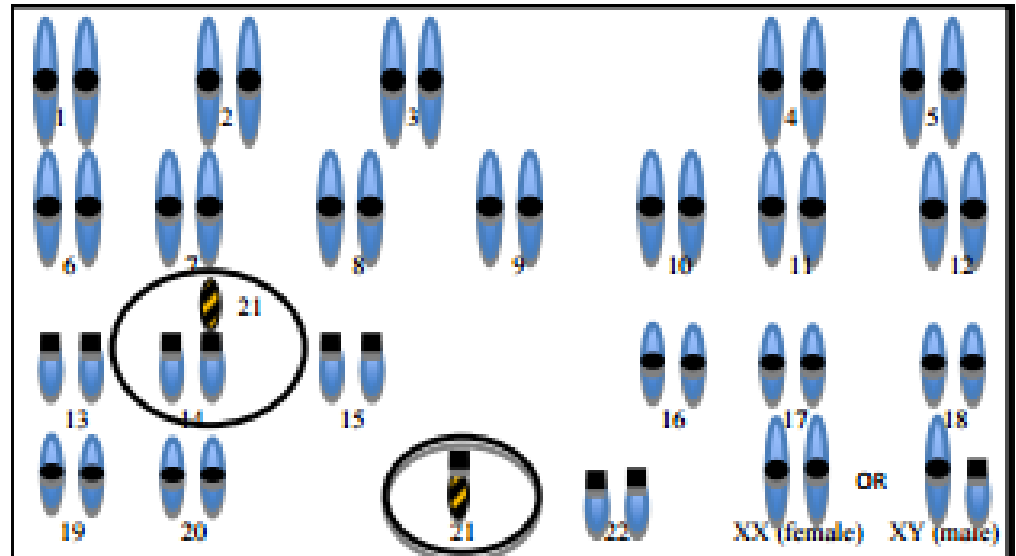


A type of balanced translocation

Can involve any of the acrocentric chromosomes: 13, 14, 15, 21, or 22

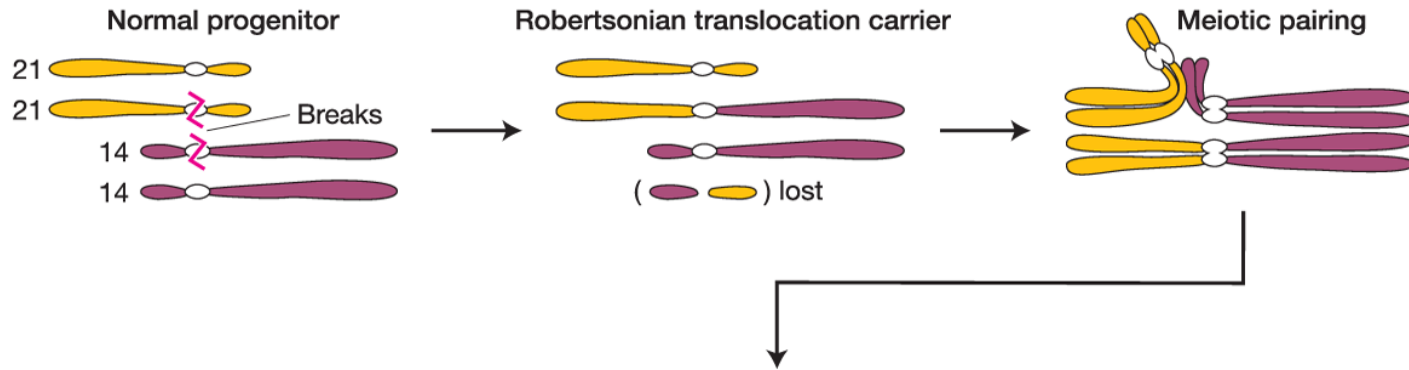
Frequency: 1/1000 babies

Phenotype usually normal



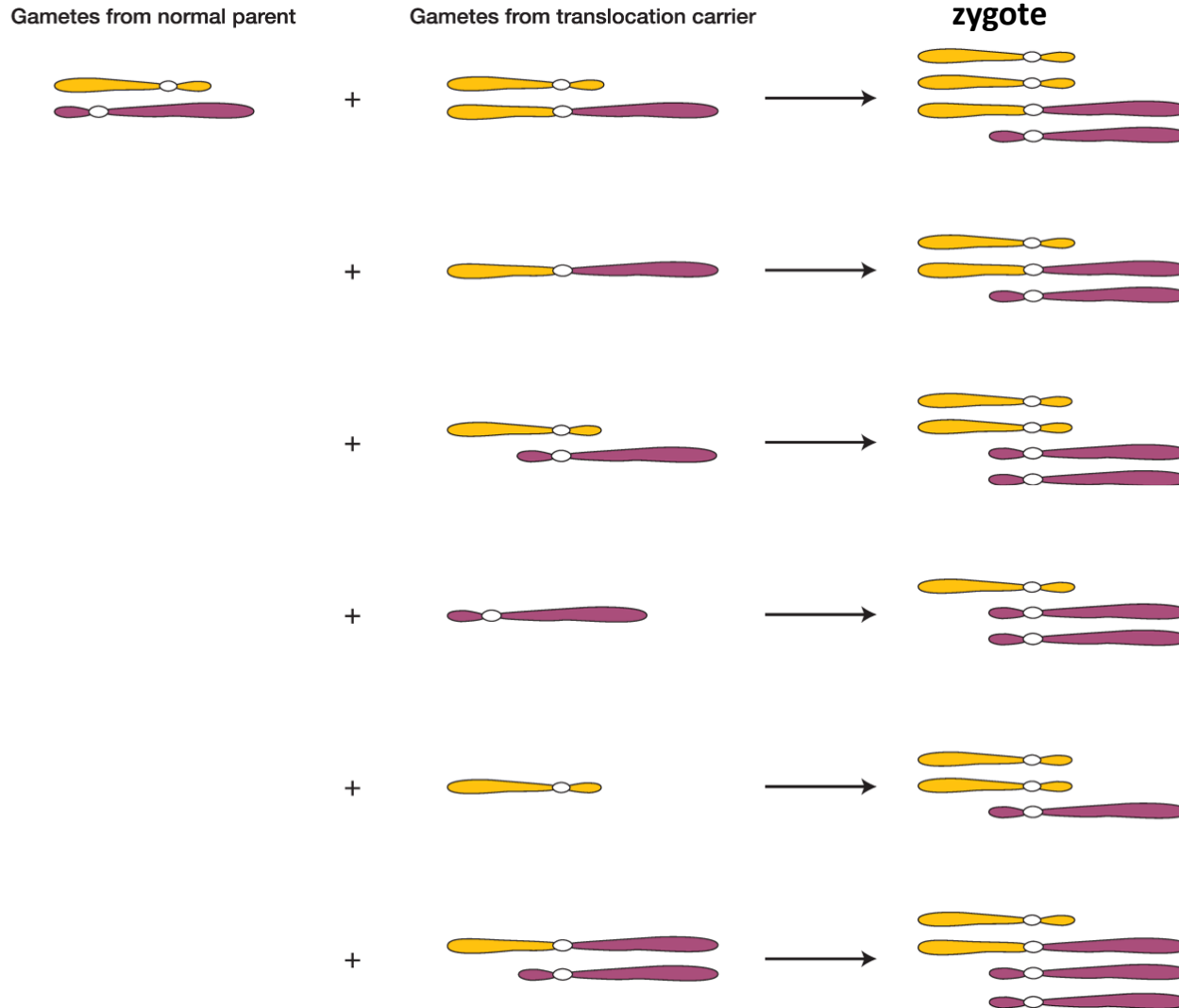
Think Break

A balanced Robertsonian translocation resulting in inheritance of trisomy 21 (Down's Syndrome)



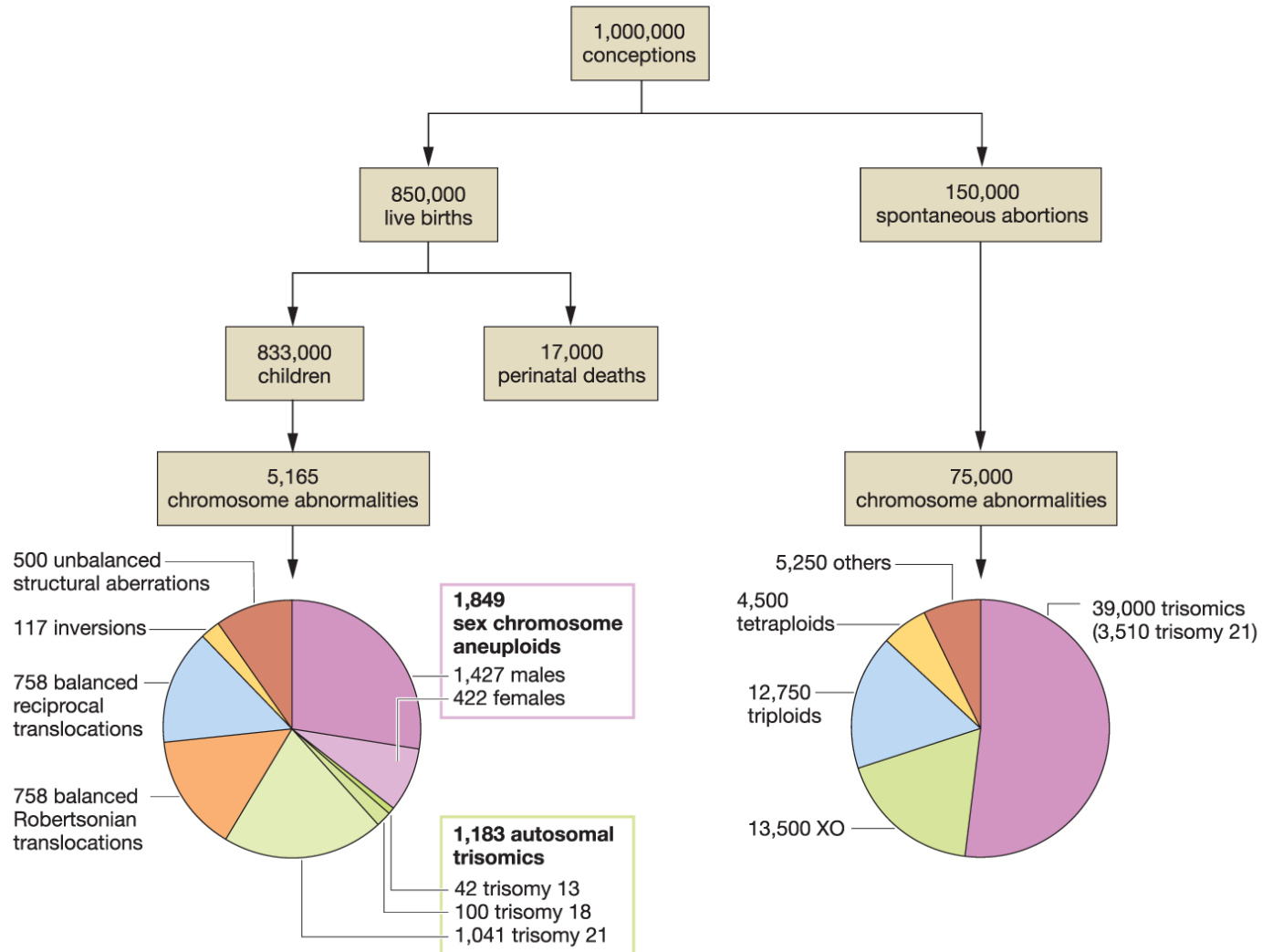
Segregation during meiosis → 6 possible gametes

A balanced Robertsonian translocation resulting in inheritance of trisomy 21 (Down Syndrome)



The fates of a million implanted human zygotes

The fates of a million implanted human zygotes



Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

The Dynamic Genome

- Discovery of transposable elements through study of “unstable” alleles
- Impact of transposable elements on the genome
- Genomic surveillance mechanisms that limit transposon mobility

Discovery of transposable elements through study of “unstable” alleles

Maize as a system for studying inheritance of traits

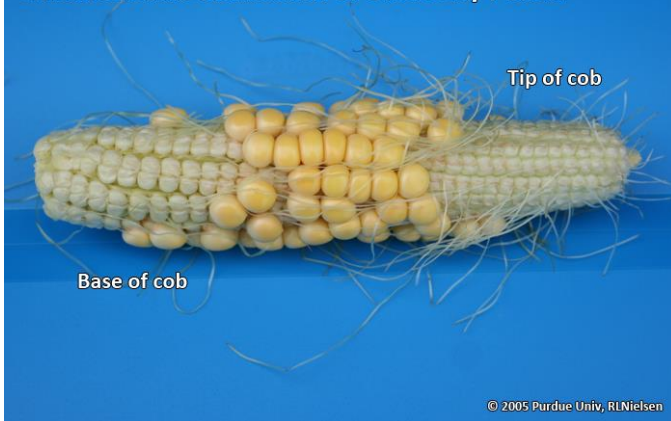
pigment genes



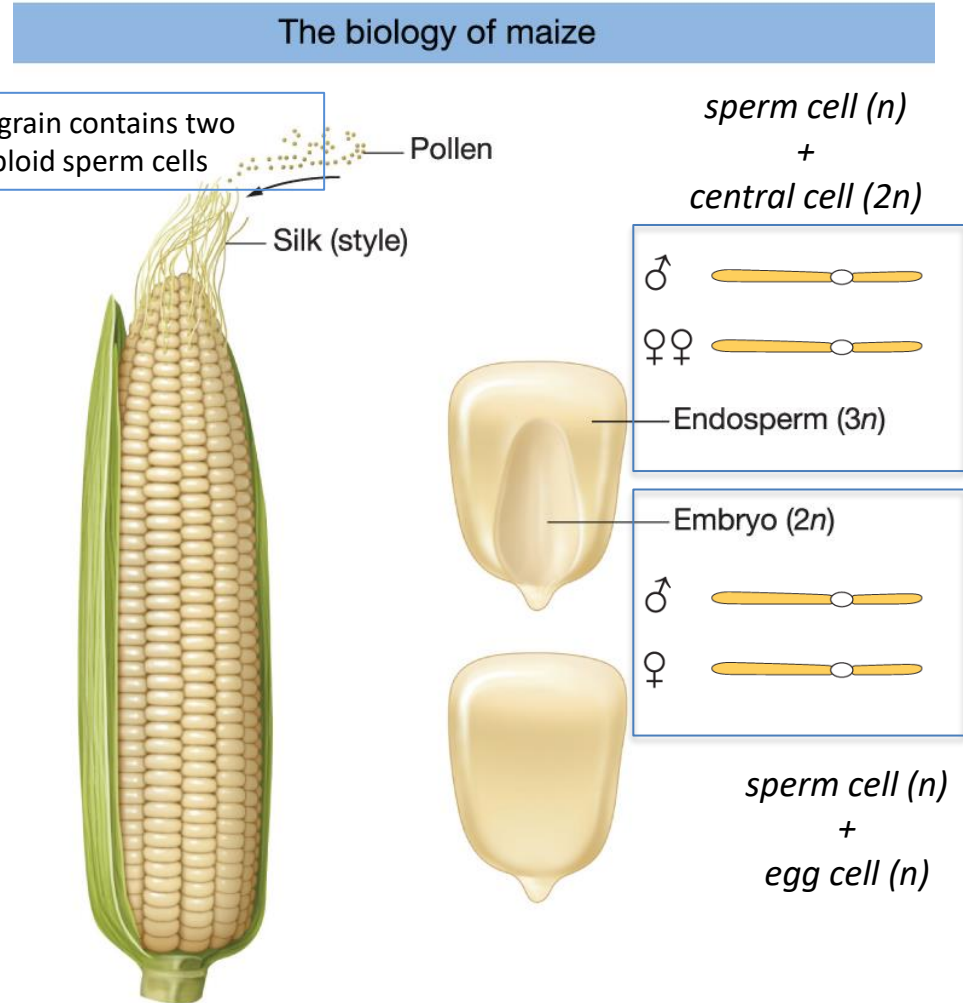
Department of Plant Breeding, Cornell University

Barbara McClintock and colleagues, 1929

Poor Kernel Set Due to
Unsuccessful Fertilization of Ovules by Pollen



© 2005 Purdue Univ, R.Nielsen



Maize as a system for studying inheritance of traits: pigment genes

The *coloured aleurone* (C) gene is important for the production of purple pigment in maize.

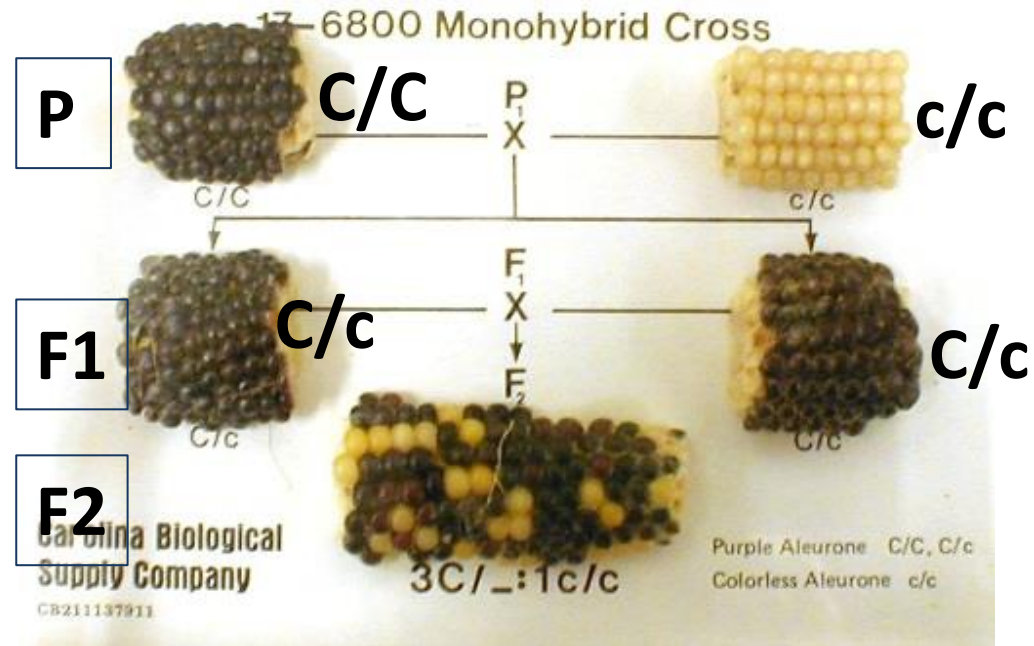
c is a recessive allele that is unable to make pigment.

C^1 is a dominant inhibitor allele that represses pigment production



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Inheritance of traits is easy to see



Normal 3:1 phenotypic ratio in the F_2

The order of dominance is $C^1 > C > c$.

Maize as a system for studying inheritance of traits pigment genes

The *coloured aleurone* (*C*) gene is important for the production of purple pigment in maize.

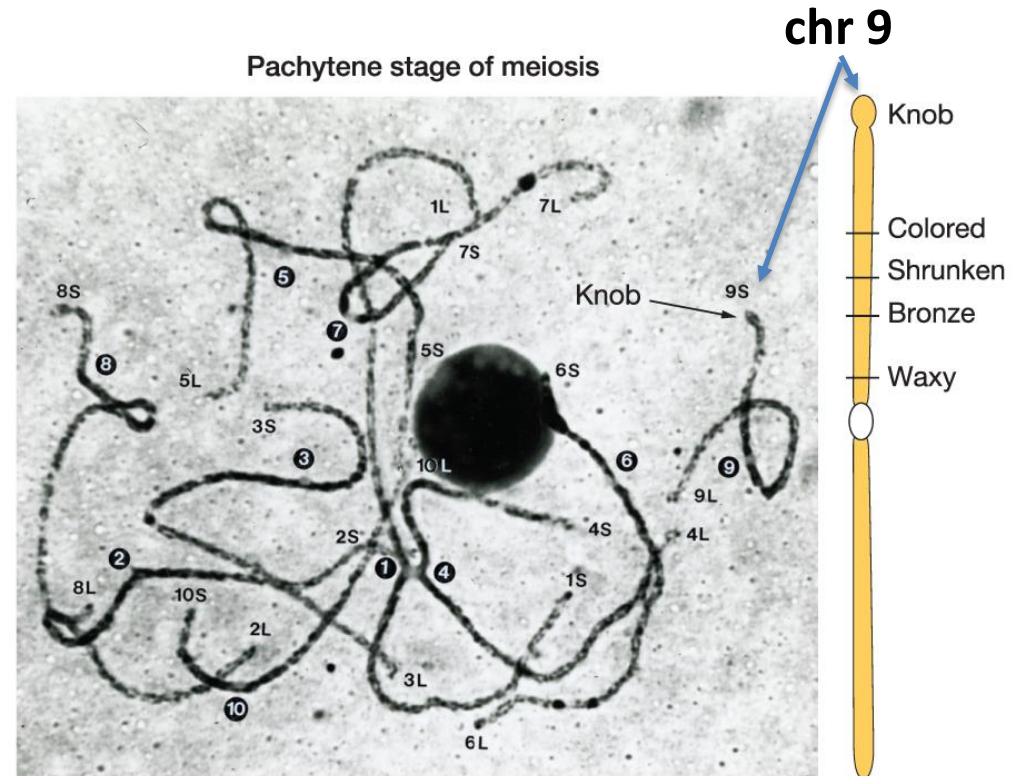
c is a recessive allele that is unable to make pigment.

C¹ is a dominant inhibitor allele that represses pigment production



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Established cytogenetic techniques



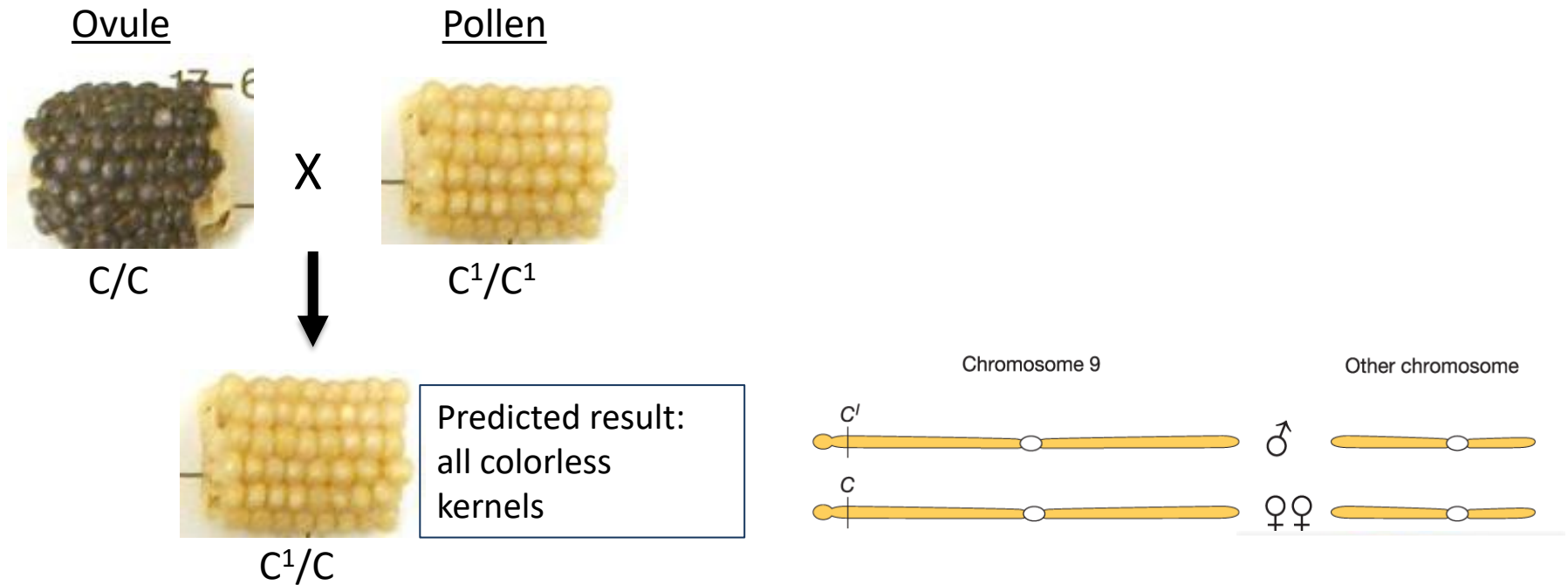
Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

Part (b) Photo kindly supplied by Ron Phillips, from a photo taken by John Stout. William Sheridan, University of North Dakota.

The order of dominance is $C^1 > C > c$.

Careful observation of “unstable” pigment gene alleles led to the discovery of transposable elements

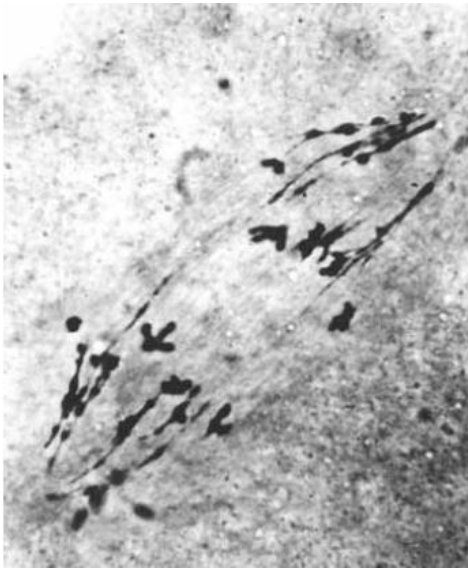
Experiment: use pollen from a C^1 (dominant negative allele) strain to fertilize C/C ovules



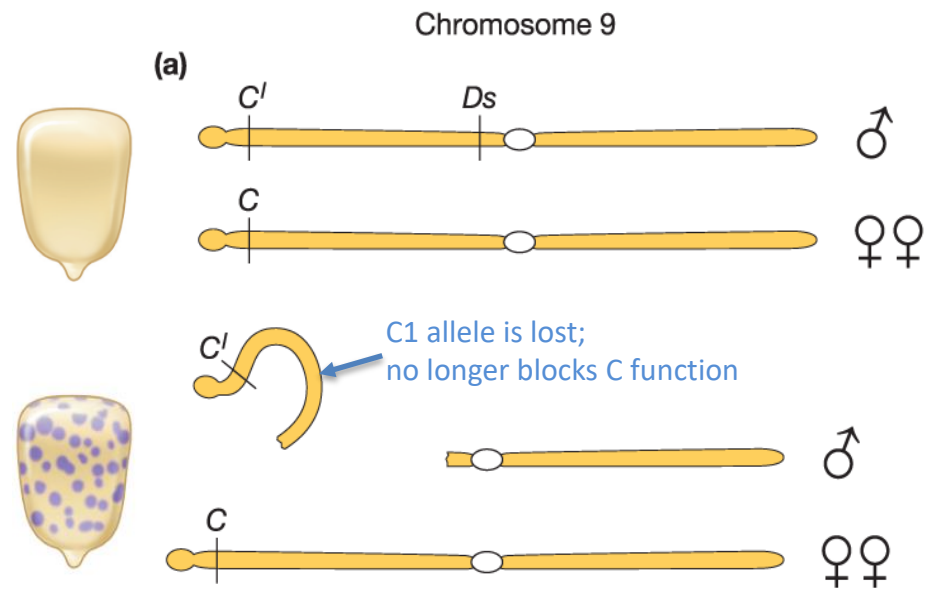
unexpected result from one particular
“pollen parent”:
some kernels have blue sectors!
How can this be?

Careful observation of “unstable” pigment gene alleles led to the discovery of transposable elements

Experiment: look for breakage in chromosomes of the pollen parent
Result: high rate of breakage in this strain

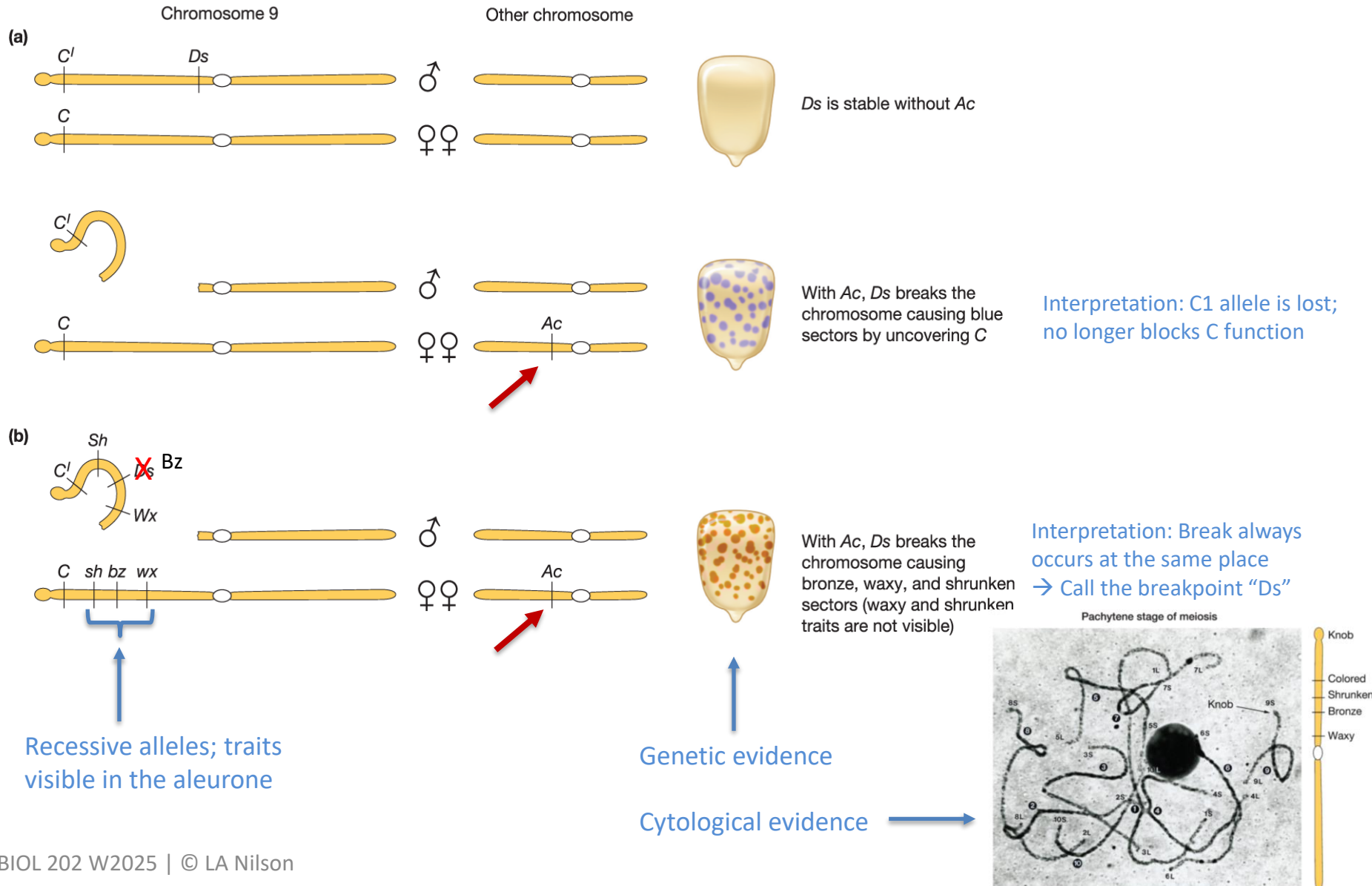


Hypothesis 1: C1 allele is lost in some cells of the developing endosperm



Interpretation: loss of the C1 allele is caused by chromosome breakage.

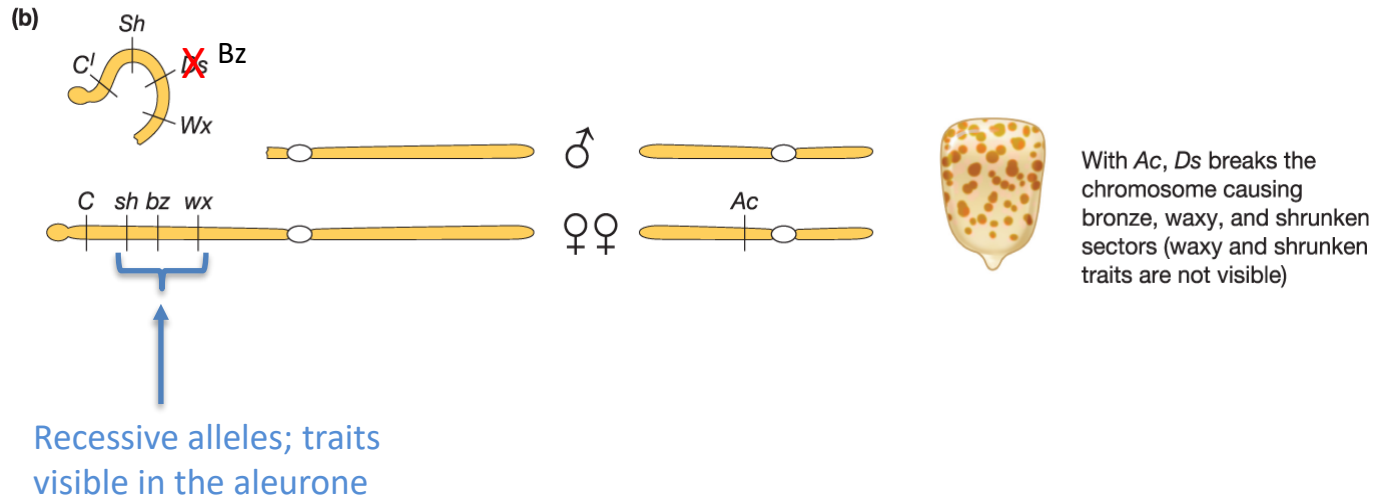
Observed genetic instability is due to chromosome breaks at specific locations



Think Break

Observed genetic instability is due to chromosome breaks at specific locations

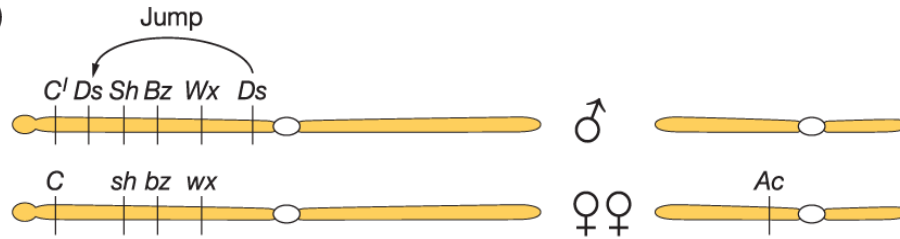
How can we describe what's happening in these sectors?



Evidence for movement of these genetic elements:

Breaks occur in a different location in a related strain

(c)

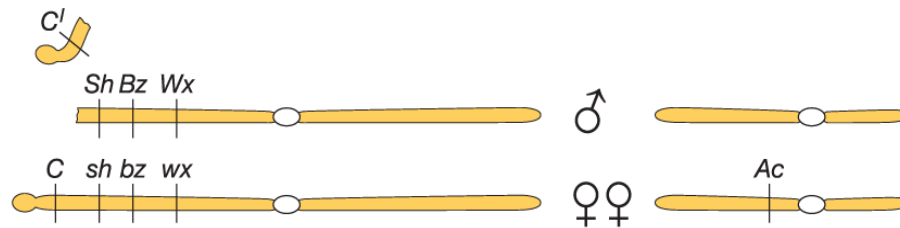


↗



With Ac, Ds excises and jumps to a new location

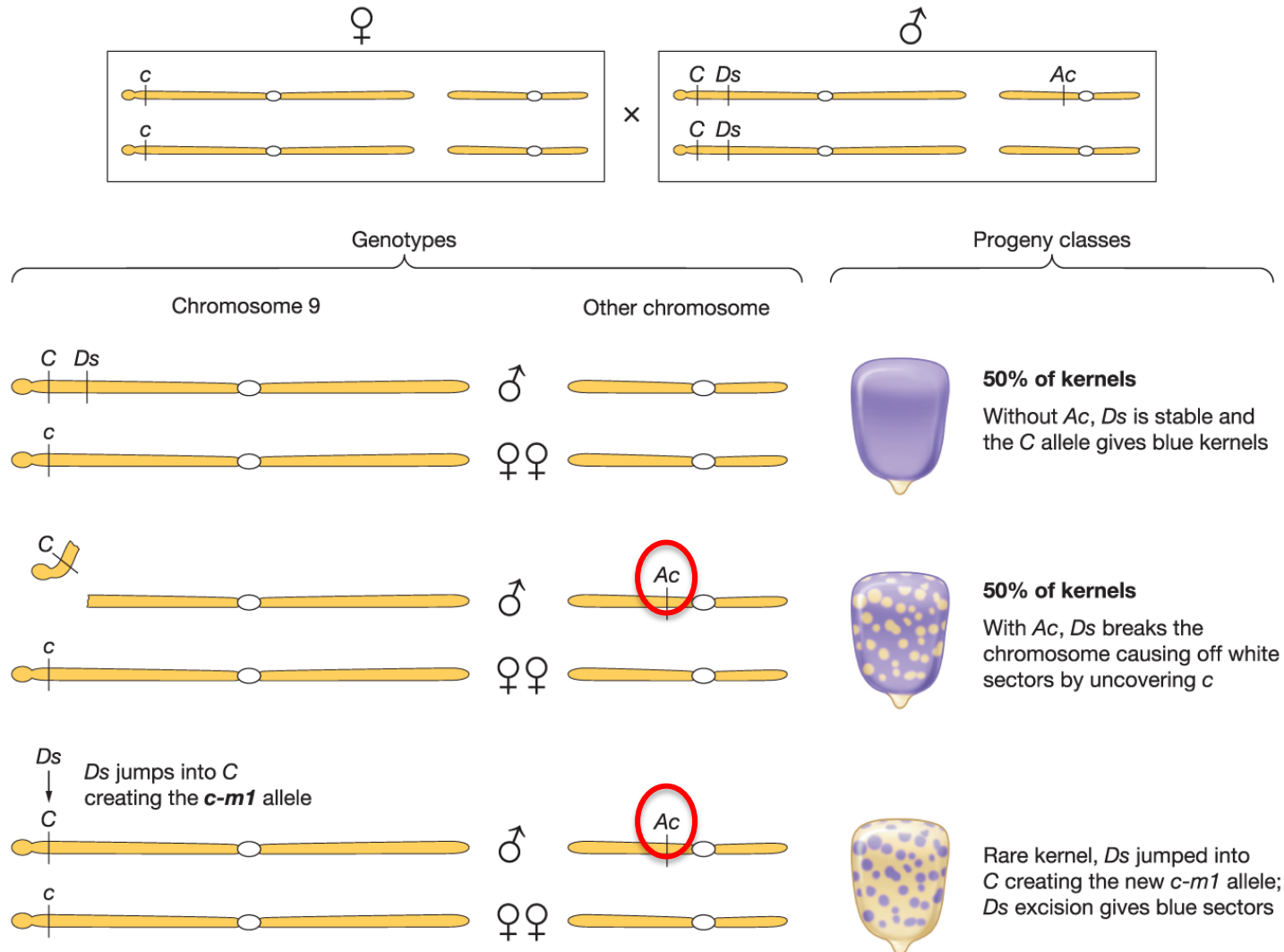
Interpretation: In some strains, Ds has moved to a new location. This new location can be inferred from the phenotype of the sectors.



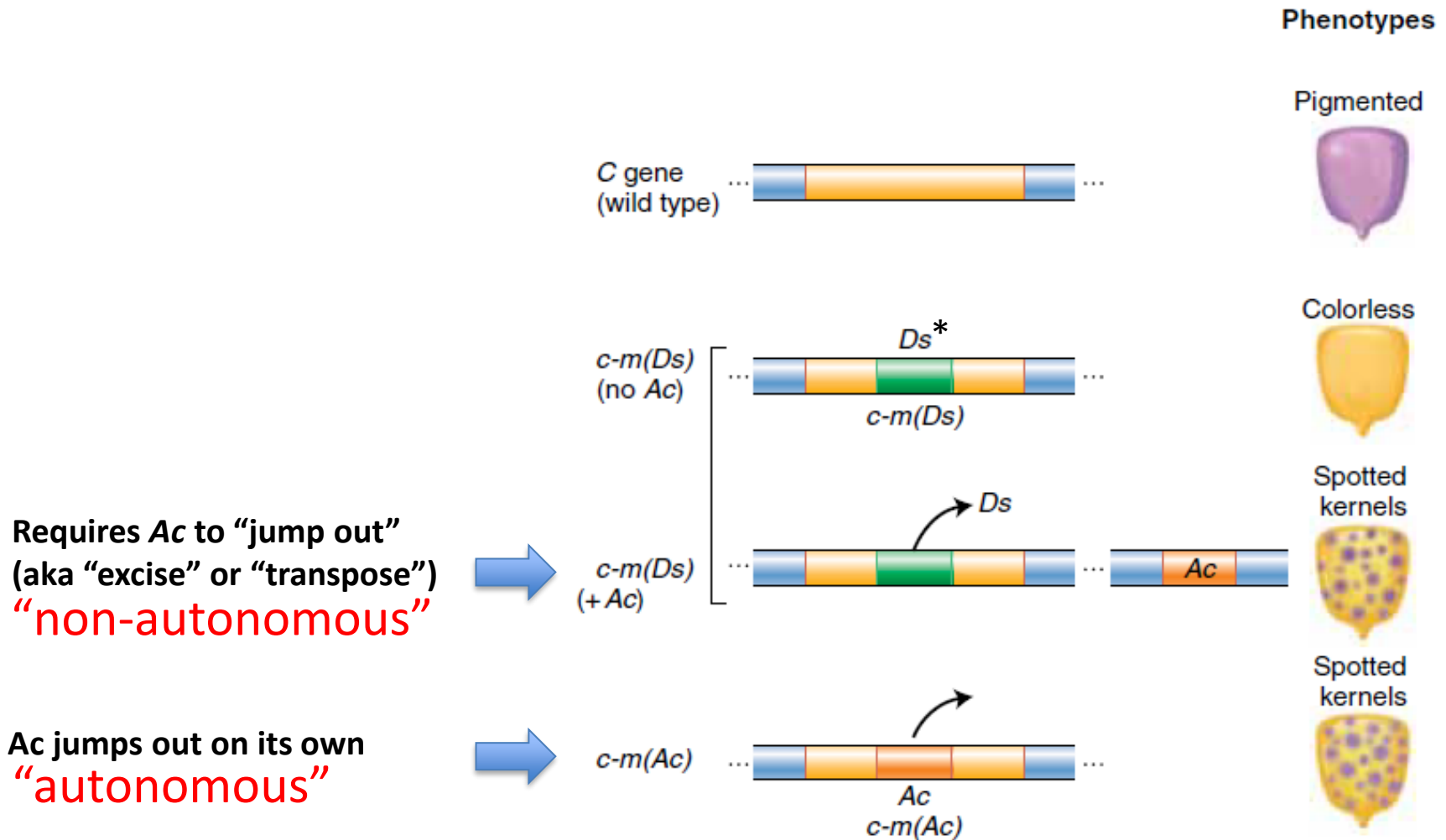
With Ac, Ds breaks the chromosome causing blue sectors with normal starch

Observation: in these sectors, pigment is restored but other traits remain unchanged

Evidence for movement of these genetic elements: multiple “jumps”, and inability to map Ac element

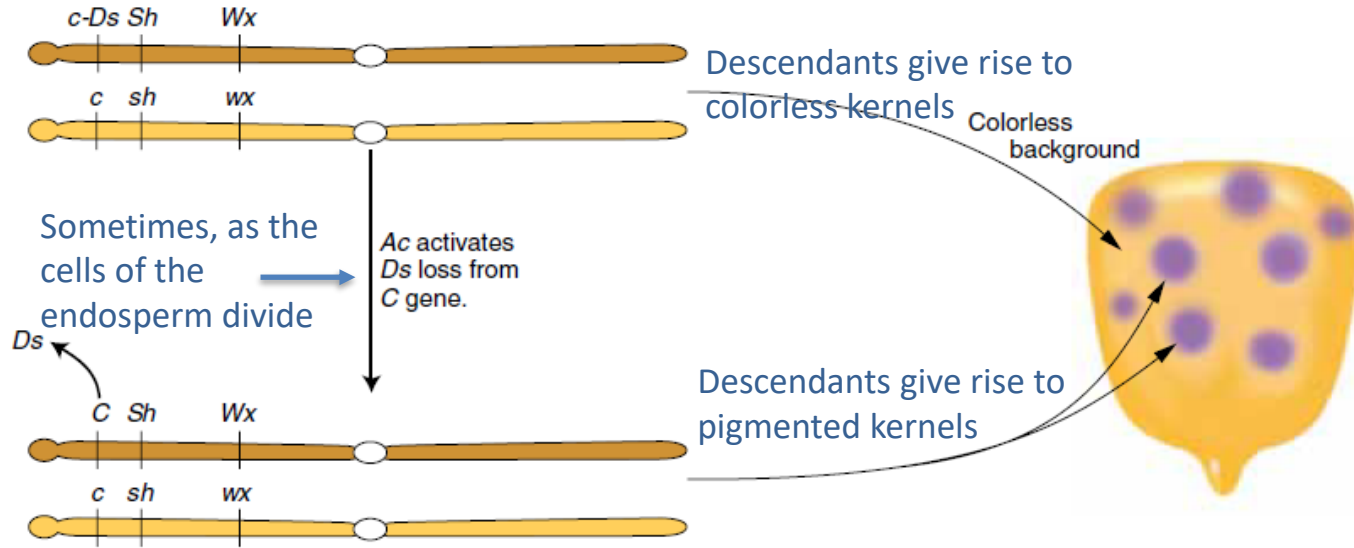


Transposable elements can be non-autonomous (like Ds) or autonomous (like Ac)

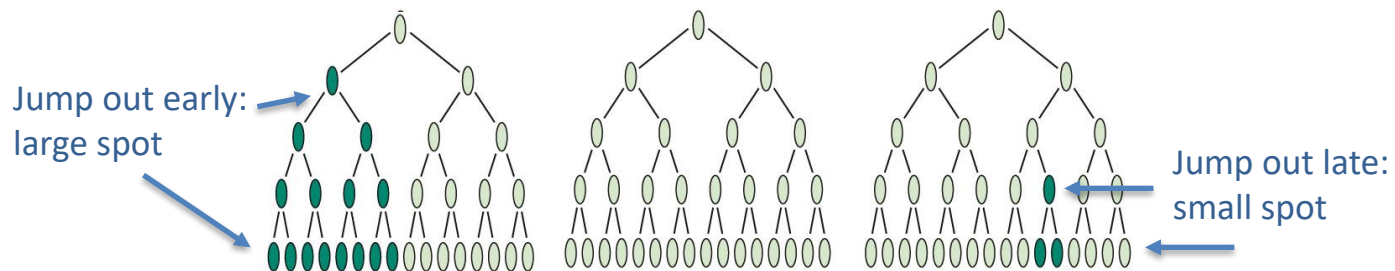


**The type of Ds is found in the c-Ds allele has a different structure. When the Ds in c-Ds excises, the two ends of the chromosome are joined together, restoring a functional dominant C allele.*

Spot size depends on when, during endosperm development, the C gene became active



Spot size depends on when in development the Ds element jumped out and restored C function.



Transposable elements

McClintock had discovered transposons, also known as “transposable elements”.

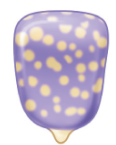
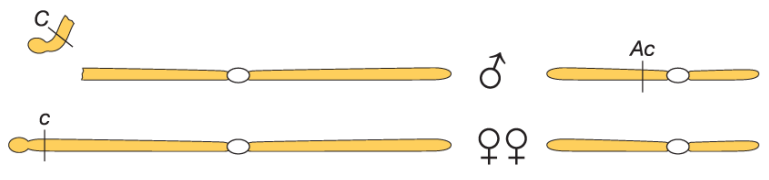
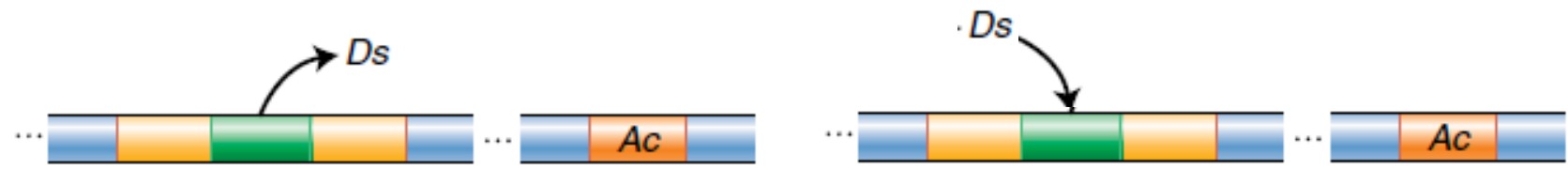
“From examination of instability of genic action at a number of known loci in maize, it is concluded that mutations need not express changes in genes, but may be the result of changes affecting the control of genic action.”

Barbara McClintock 1953

Think Break

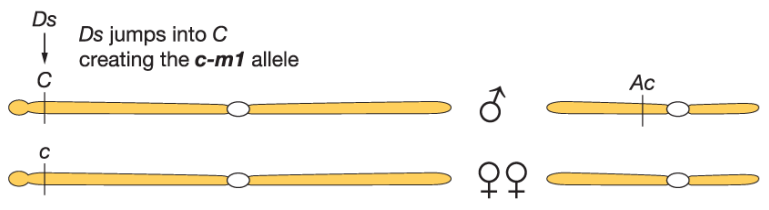
Reflection

In the presence of Ac, Ds mobilization reverts the c(Ds) mutant allele frequently. But in rare cases Ds mutates the C allele. What happens, and why the difference in frequency?



50% of kernels
With Ac, Ds breaks the chromosome causing off white sectors by uncovering c

Jumping out: frequent



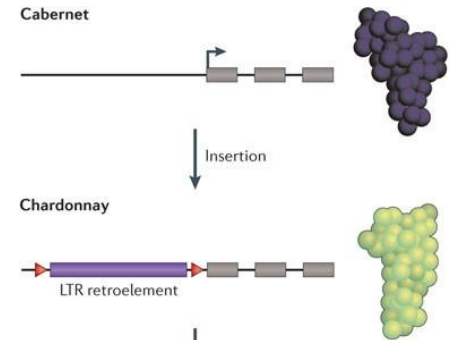
Rare kernel, Ds jumped into C creating the new c-m1 allele; Ds excision gives blue sectors

Jumping in: rare

Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

Transposable elements are found in all organisms

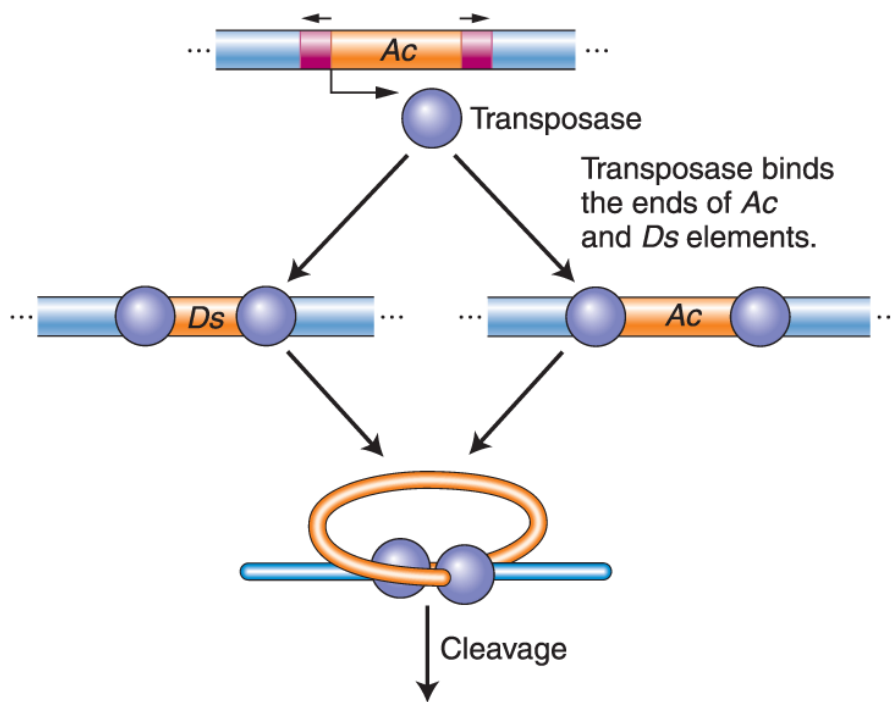
- Transposable elements are everywhere. Not just a weird plant phenomenon!
- Studies in maize identified transposable elements genetically, but it was later studies in yeast and *Drosophila* that identified the actual pieces of DNA that are mobile and discovered how they move.
- Transposons, or “transposable elements” are ancient, found in all organisms: bacteria, plants, yeast, multicellular animals.
- Transposable elements fall into two classes:
 - Class 1: Retrotransposons (eukaryotes)
 - Class 2: DNA transposons (prokaryotes and eukaryotes)



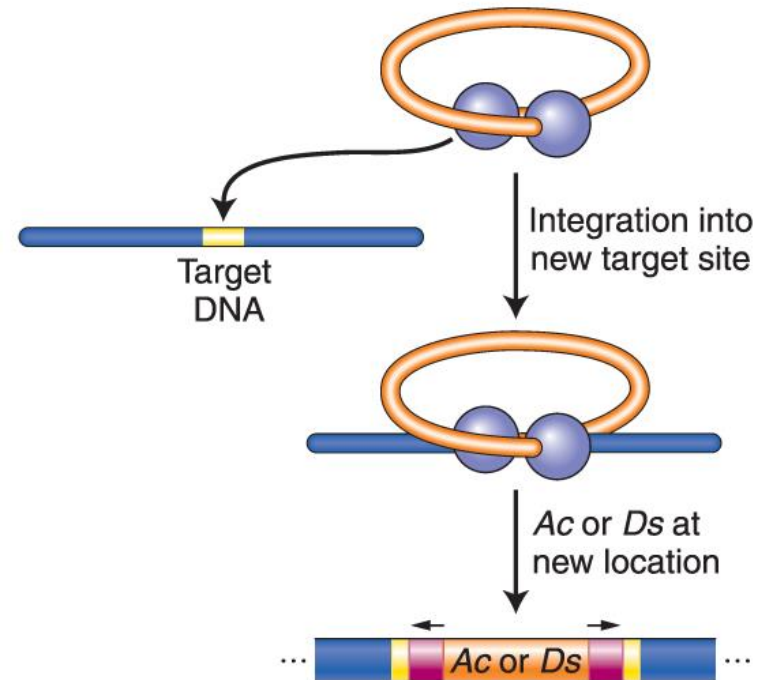
Class 2: DNA transposons

Example: the Ds and Ac elements in maize

Activator transposase catalyzes excision and integration

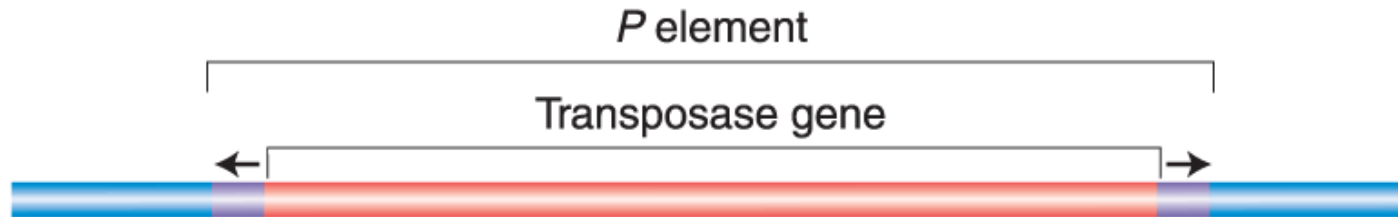


There are multiple transposon families in maize, each with autonomous and non-autonomous elements. *Ac* and *Ds* are in the same family. Each has a transposase version that is specific for its own family.



Class 2: DNA transposons

Example: P-element in *Drosophila*



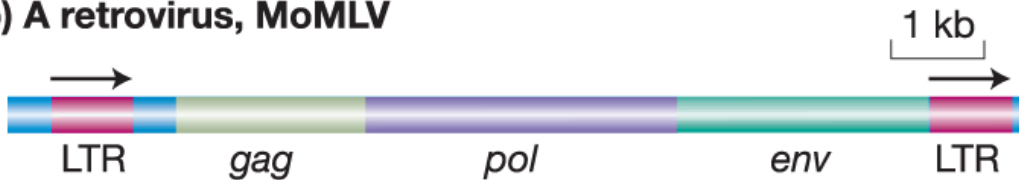
- Encodes a single protein: the transposase
- 31-bp inverted repeats flank the transposase gene
- Resembles transposons found in bacteria

Class 1: Retrotransposons

(a) *Ty1* in yeast



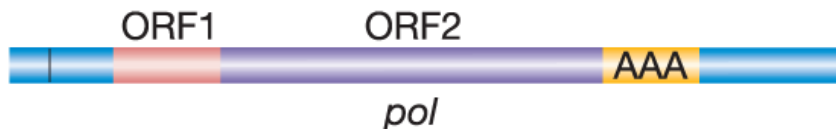
(b) A retrovirus, MoMLV



(c) *Copia* in *Drosophila*

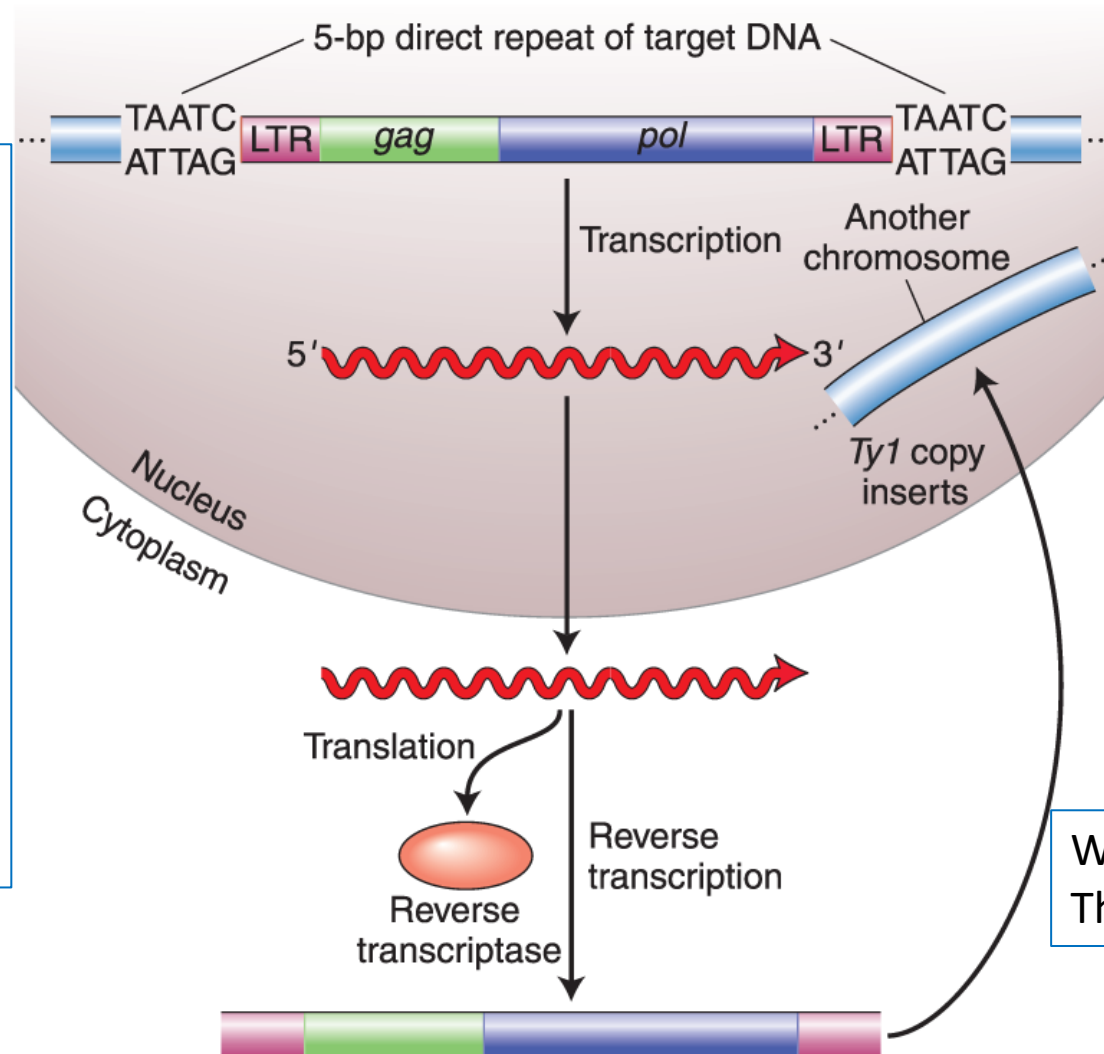


(d) L1, a human LINE



- Similar to retroviruses
- First identified in yeast as a repeated sequence in the genome.
- Found in many organisms
- LINE (long interspersed element) is a retrotransposon found in the human genome.

Retrotransposons move through an RNA intermediate



What is the impact of these
transposable elements on the
genome?